

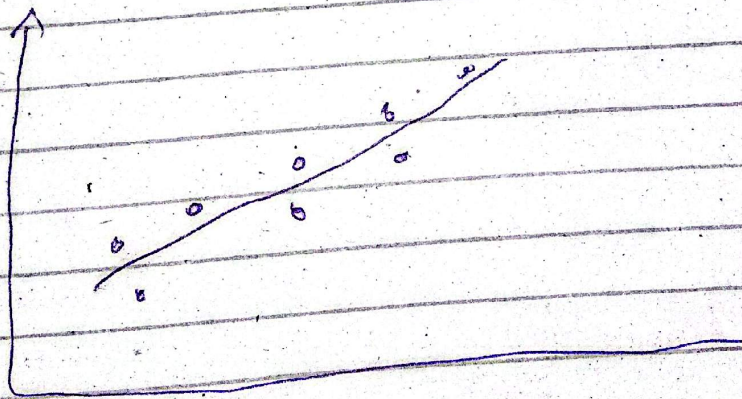
→ How to draw the best fit line in linear regression:
using least square method!

Given that:

1. Write a linear equation that "best-fits" the data in the table shown below:

x	1	2	3	4	5	6	7
y	1.5	3.8	6.7	9.0	11.2	13.6	16

Like example: we will draw a line which are best fit. means all the data points will be near to that line.



So this best fit line we will draw using least square method

step Draw the table.

X	Y	XY	X ²
1	1.5	1.5	1
2	3.8	7.6	4
3	6.7	20.1	9
4	9.0	36	16
5	11.2	56	25
6	13.6	81.6	36
7	16	112	49
$\Sigma x = 28$	$\Sigma y = 60.8$	$\Sigma xy = 314.8$	$\Sigma x^2 = 140$

$$\text{Linear eq} = \boxed{y = mx + b}$$

$$\text{or } y = b_0 + b_1 x$$

$$\text{slope} = m = b_1$$

$$y\text{-int} = b = b_0$$

$$\text{so } y = mx + b$$

step 1: calculate slope.

$$m = \frac{n \Sigma xy - \Sigma x \Sigma y}{n \Sigma x^2 - (\Sigma x)^2} = \frac{7(314.8) - (28)(60.8)}{7(140) - (28)^2}$$

$$m = \frac{473.2}{196} = \boxed{2.414}$$

$$\text{so } \boxed{m = 2.4142857}$$

Now let's calculate the y-intercept which is b .

$$b = \frac{\sum y - m \sum x}{n} = \frac{61.8 - (2.4142857)(28)}{7}$$

$$[b = -0.828571]$$

so

$$[y = 2.41x - 0.83]$$

so let's put the value:

$$y = 2.41(2) - 0.83 = [3.99]$$

so when we put value $n=2$ as a x we get 3.99 which is close to 3.8.

Now another value from the table:

so let's put the value 5.

$$y = 2.41(5) - 0.83 = [11.22]$$

so again the 11.22 is very close to 11.2.

now $n=7$:

$$y = 2.41(7) - 0.83 = [16.04]$$

so 16.04 is very close to

16. so we can say

that the $y = 2.41x - 0.83$ equation fits the observed data from table very well.

n Value	Actuals (scatter Dots)	Predicted $y = 2.041n - 0.83$ (line points)
1	1.05	1.58
2	3.08	3.99
3	6.07	6.40
4	9.00	8.81
5	11.02	11.22
6	13.06	13.63
7	16.00	16.04

Now place a straight ruler through these predicted (eg from $n=1, y=1.58$ to $n=7, y=16.04$) and draw a solid straight line across your graph.

Scatter plot:-

